

Master production schedule (MPS) or Production Plan

Master production schedule (MPS) is the second input that goes into the MRP plan after the bills of material (BOM). The MPS details how many end items will be produced in a given period of time.

To develop a master production schedule, there are two steps involved.

- 1). Calculate the projected on-hand inventory at the end of a period
- 2). Determining the timing and size of the production quantities of specific products

Step 1 is to calculate the projected on-hand inventory at the end of a period. So, if you have inventory in your stock then you don't need to produce anything because that inventory is good enough to meet the demand. So first we have to find out what the inventory is at the end of a period. Next is determining the timing and size of production quantities of specific products. Now if we determine that there is less inventory and demand is going to exceed in a certain period, then that means that we will have to plan the production. And in order to plan the production, we will need the size of production, that is how many items do we require, and the timing, that is when is it required to be produced.

Calculate the projected on-hand inventory at the end of a period

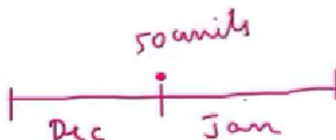
So now let's look at each of these steps in detail. Now the first step is calculate the projected on-hand inventory at the end of a period. This is the amount of inventory available after the demand has been satisfied. So let us look at the formula.

Projected on-hand inventory at the end of this period = (on-hand inventory in last period) + (MPS quantity due this period) – (projected requirements this period)

So, if we look at this formula, what this says: Projected on-hand inventory at the end of this period, that means whatever period is in consideration, is equal to on-hand inventory in last period plus the MPS quantity due this period, that means what is the quantity which is being manufactured and which is going to be available in the current period, minus the projected requirements this period. Projected requirements means what is going to be the demand.

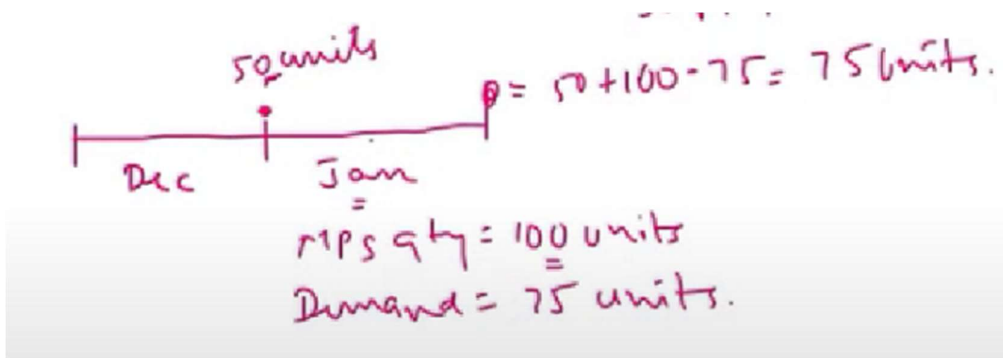
If you look at this formula, what this basically means, you are taking the inventory which is available at the beginning of this period or at the end of last period, adding it to the MPS quantity that you are going to receive in this period. So, **(on-hand inventory in last period) + (MPS quantity due this period)** together is your **supply** available. And then you are subtracting the projected requirement this period. That means how much do we need to ship to the customer? So, **(projected requirements this period)** is your **demand**. So, in a sense, you are taking all the supply available which can give you the inventory for that period, and then you are subtracting the demand. And that will give you the **projected on-hand inventory at the end of this period**.

Now, let's say we are looking at calculating the inventory at the end of Jan. So let's say you have this period here.



Let's say this is December, and then you have this as Jan. So first we need to find out the on-hand inventory of last period. So here at the end of December and beginning of Jan, what is the on-hand inventory? Let's say this is 50 units. So this is 50 units. Now there could be some quantity being manufactured as part of the MPS plan. So, we need to find out if there is any quantity being delivered in January. So, this 50 units is of course available in January which is the on-hand inventory at the end of December. Now if there is some other supply which is going to get delivered in the month of January it could be the MPS. So let's say that we are planned to receive 100 units as part of the MPS planning. So, in January your MPS quantity is equal to 100 units. So now the total supply is this 50 here and this 100 here. So, this on-hand inventory in the last period this is 50 and MPS quantity due this period this is 100. Now next we need to find out how much quantity is going to be shipped to the customers. So, let's say projected requirements this period is 75. So, your supply is 150 and your demand is equal to 75 units. So if we now calculate the projected inventory at the end of January, this will be equal to 50 which is this plus 100 minus 75. So basically 150 minus 75 which is 75 units.

$$\text{(Projected On-hand inventory at the end of this period)} = \text{(On-hand inventory in last period)} + \text{(MPS quantity due this period)} - \text{(Projected requirements this period)}$$



Now with this calculation there are certain rules. So let's look at one such rule.

Rule # For the projected requirements for this period, we should use the larger quantity known between the forecast and customer orders booked.
This helps in avoiding any stock out situations.

Now this rule says that for the projected requirements, we should use the larger quantity known between the forecast and the customer orders booked. So basically, as we mentioned, this is the demand. Now demand can be two. One is the forecasted demand where you are projecting what the actual demand is going to be. And then there is the actual customer orders which are booked. What this rule is saying is that if let's say forecast is 100 and customer orders booked is 60, so this rule is saying is that for this demand we should use the larger quantity known between forecast and customer orders booked; whichever is the larger one. So if 100 is the larger one we should be considering 100 as the projected requirement for this period. Now this is basically done to avoid any stock-out situations. Let's say, you assume that the demand is 60, but let's say you are only in this period at that point in time, and towards the last week you again get a quantity of 40. So, what that will mean is that you are not able to supply 40 because you only projected 60 and you planned for 60. So that will be a stock-out situation and you will lose money. So, to be on the safer side, we will use the larger of the two. If let's say the forecast is 100 and if the customer orders booked is 160, then we will use 160. So this is one of the rules which we have to take into consideration.

Example 1

Now let us look at another example.

Item: Ladder back chair		
On hand: 55	April	
	Wk:1	Wk:2
Forecast	30	
Customer Orders booked	38	
MPS qty	0	
Projected on-hand qty		

Step 1

Now in this example, the period is weekly. Let's say the manufacturer of the ladder- back chair - so this is one of the type of chairs that the manufacturer produces. And the manufacturer of this chair produces the chair to stock and needs to develop an MPS for it. Now we are saying that this guy is manufacturing to stock. Now marketing has forecasted a demand of 30 chairs. So forecast here is 30. That means they are saying that in week one the demand will be 30 as per the marketing team. But the actual orders booked are 38. So 38 are the actual orders booked. Now the current on-hand inventory is 55. That means, as we had seen earlier, when we start week one you already have 55 ladder back chairs available in your inventory. And as per the plan there is no MPS quantity which is due to be received in this week.

So we need to now calculate the projected on-hand quantity at the end of week one. Let's use this formula here. So projected on-hand quantity at the end of the period is equal to on-hand inventory in last period which is 55. So this becomes - so 55 plus

the MPS due this period which is 0. So plus 0 minus the projected requirements this period which will be 38. Because as we saw this rule here we have to take the larger between the forecast and the customer orders booked. So this is 38. So minus 38. So this will be equal to 17. So 17 chairs is the projected on-hand quantity at the end of week one.

Now let's similarly look at week 2. So let's say the forecast was 30 units, and the customer orders booked is 27 units, and MPS quantity, as you know, is 0, let's say. You are not aware of any MPS quantity to be received. Now what is the projected on-hand? So at the end of week 1 you had 17 units. So on-hand inventory in last period, this is 17. Plus the MPS quantity due this period which is 0. So plus 0 minus the projected requirements this period. So we have to take the higher of these two. So minus 30. So this will be equal to minus 13. So as you can see, the demand is higher than the supply and there is no MPS quantity which is coming to rescue us. So this means that we will not be able to cater to the entire demand. We will have an opportunity lost of 13 units. So this negative sign shows shortage. And hence an MPS quantity needs to be made available in order to avoid the stock-out.

Item: Ladder back chair		
On hand: 55	April	
	Wk:1	Wk:2
Forecast	⇒ 30	30
Customer Orders booked	→ 38	27
MPS qty	→ 0	0
Projected on-hand qty	55 + 0 - 38 = 17	17 + 0 - 30 = -13

needs to be provided

So, this here needs to be provided. So this now leads to our step number two which is determining the timing and size of the production quantities of specific products. So let's look at Step 2 now.

Step 2

2) Determining the timing and size of the production quantities of specific products

Item: Ladder back chair		Order policy: 150 units; Lead time: 1 week							
On hand: 55	April				May				
	Wk:1	Wk:2	Wk:3	Wk:4	Wk:5	Wk:6	Wk:7	Wk:8	
Forecast	30	30	30	30	35	35	35	35	
Customer Orders booked	38	27	24	8					
MPS qty	0								
Projected on-hand qty									
MPS – start of production									

Now, as you can see here, I have created this table. You would call it the MPS table. And we are going to determine the timing and size of production quantities for specific products. So we had seen in week one we had 30 forecast, 38 as customer orders booked, 0 MPS, and projected on-hand quantity was 17. Now in April week 2, we saw the demand as 30, the on-hand was 17, and there was no MPS. So 17 minus 30 - in this case we are doing supply minus the demand - and so the negative 13 is the answer which means that there is a shortage. So we don't want to have shortage. So what we would do is we need MPS quantity here. We need some supply to be made available. That means there has to be some production done.

Now what has been given here is that the order policy is for 150 units. So basically, you know, it is most optimal for your production facility to produce a lot size of 150 units. So every time you run the production line you will have to produce 150 units. And the lead time is one week. That means if you want the material to be available in week two, then it has to be produced in week one. So one week - okay it could be two weeks, that means then this would have to be two weeks prior we would have to schedule the MPS production start. So basically what this means is that we will now have a start of production of 150 units. So what this means is that you will now have an MPS quantity available of 150 here. So now projected on-hand will become on-hand inventory in last period which was 17. So 17 plus MPS quantity which is 150, minus the projected requirements which is 30. So this is 137. So 137 is basically your on-hand inventory at the end of week two. Okay.

Now we don't know what needs to be produced here until we know the situation in week three. Now in week three we have been given a forecast of 30 and customer orders booked of 24. We do not have an MPS quantity so we'll assume this is 0. So now the projected on-hand becomes on-hand inventory in past period, so which is 137, plus the MPS quantity which is 0, minus the projected requirement which is 30, the higher of the two, so 30. So 137 minus 30 which is 107. So projected on-hand becomes 107.

Now let's go to week four. So again MPS is zero. Now the calculation becomes the on-hand inventory in last period which is 107 plus the MPS quantity due this period which is 0 minus the projected requirements this period which is 30. So 107 minus 30 which is 77. So the projected on hand at the end of week 4 is 77. So we are able to cater to the demand with the on hand that we were able to get because of this lot size of 150. So we don't need to produce anything till week four.

Now let's see week five. So here, you know, the customer orders booked are not given because it's too far away for now. So what will happen is that the demand will be the forecast. So let's see the projected on-hand at the end of week five will be on-hand inventory in last period which is 77, plus zero, minus the projected requirements this period which is 35. So 77 minus 35 which is 42.

So now at the end of week 6, the projected on-hand quantity will become 42 minus 35, which is 7.

At the end of week 7, however, now this is reducing, right? So at the end of week 7, it'll be on-hand which is 7, plus the MPS which is right now 0, minus the projected requirement which is 35. So this will end up in a negative number. So what this means is that we need an MPS quantity here. Now this negative number is going to be minus 28. So what that means is that we will be fine if we produce only one lot of 150 units. Again the lead time is one week, so if we have to have the supply in week seven, we need to get the units in week six. So here we will produce another lot of 150 units. So then the MPS quantity will become 150. So now my projected on-hand becomes 7, which is the end of week six, plus MPS which is 150, minus the demand which is 35. So this becomes 157 minus 35 which is 122 units. So projected on-hand becomes 122.

Item: Ladder back chair		Order policy: 150 units; Lead time: 1 week							
On hand: 55	April				May				
	Wk:1	Wk:2	Wk:3	Wk:4	Wk:5	Wk:6	Wk:7	Wk:8	
Forecast	30	30	30	30	35	35	35	35	
Customer Orders booked	38	27	24	8					
MPS qty	0	150					150		
Projected on-hand qty	17	107	77	42	7	122	87		
MPS - start of production	150					150			

$$\begin{aligned}
 &7 + 150 - 35 \\
 &= 157 - 35 \\
 &= 122
 \end{aligned}$$

$$\begin{aligned}
 &107 + 0 - 30 \\
 &= 77
 \end{aligned}$$

$$\begin{aligned}
 &7 + 0 - 35 \\
 &= -28
 \end{aligned}$$

Now again week 8 we will have on-hand of last period which is 122, plus MPS is 0, minus projected requirement this week is 35. So basically you will get 87. So this now gives you the plan that we need to start producing 150 units in week 1, and 150 units in week 6. So this is how a typical Master Production Schedule planning is done.